PHYSICAL CHEMISTRY

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Topic : Chemical Equilibrium & Ionic Equilibrium (Question Bank # 02)

1. Ans. (B)

Sol.

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

Millimoles in 100 ml buffer	2	0.5	1
	1.5	x	1.5

For acidic buffer

$$pH = pKa + \log\left(\frac{5}{a}\right)$$

$$= 5 - \log 2 + \log\left(\frac{1.5}{1.5}\right)$$

$= 4.7$

2. Ans. (B)

Sol.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

at t = 0;	1	–	–
At equilibrium	$1-\alpha$	α	α

Avg. M_w of mixture

$$\bar{M} = \frac{M}{1+\alpha}$$

But $d = \frac{P\bar{M}}{RT}$

$$d = \frac{PM}{RT(1+\alpha)} \Rightarrow 1+\alpha = \frac{PM}{dRT}$$

$$\alpha = \frac{PM}{dRT} - 1$$

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3. Ans. (B)

Sol. $Na_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

At I equilibrium: $2P$ P

At new equilibrium: $3P$ P^1

as temperature is not charged so K_P is constant.

$K_P = (2P)^2 \times P = (3P)^2 \times P^1$

$P' = \frac{4}{9}P$

$$\frac{P_{total}\,(\text{new equilibrium})}{P_{total}\,(\text{original or I equilibrium})} = \frac{\left(3P + \frac{4}{9}P\right)}{2P + P} = \frac{\frac{31}{9}P}{3P}$$

4. Ans. (C)

Sol. (A) $CH_3COOH + NH_4OH \longrightarrow CH_3COONH_4 + H_2O$

m moles	3	3	–
	x	x	3

Salt of W.A & W.B. $\Rightarrow$ Buffer

(B) $Na_2CO_3 + H_2CO_3 \longrightarrow 2NaHCO_3$

m moles	2	4	–
	x	2	4

H_2CO_3 & $NaHCO_3$ from buffer.

(C) $HCl + Na_3PO_4 \longrightarrow Na_2HPO_4 + H_2O$

3	1	–
2	x	1

$Na_2HPO_4 + HCl \longrightarrow NaH_2PO_4 + H_2O$

1	2	–
x	1	1

$NaH_2PO_4 + HCl \longrightarrow H_3PO_4 + H_2O$

1	1	–
x	x	1

Only H_3PO_4 remain, so no buffer

(D) $NH_4Cl + NaOH \longrightarrow NH_4OH + NaCl$

1	0.5	–	–
0.5	x	0.5	0.5

NH_4OH & NH_4Cl form buffer.

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5. Ans. (A)

Sol.

	$nA(g)$	$\rightleftharpoons$	An
$t = 0$; moles	1		–
at equilibrium	$1-\alpha$		$\frac{\alpha}{n}$

$$\frac{PV}{RT} = n_{total} = 1-\alpha+\frac{\alpha}{n} = 1-\left(1-\frac{1}{n}\right)\alpha$$

$$= 1-(n-1)\frac{\alpha}{n}$$

$$k_c = \frac{\left(\frac{\alpha}{nV}\right)}{\left(\frac{1-\alpha}{V}\right)^n} = \frac{\alpha}{n}\ V^{n-1} \qquad \text{(given } \alpha \text{ is very small} \Rightarrow (1-\alpha)\simeq 1)$$

$$\frac{\alpha}{n} = \frac{k_c}{V^{n-1}}$$

6. Ans. (C)

Sol. (I) $\left[Ag^+\right] = \sqrt{k_{sp_{(AgCl)}}} = 10^{-5}\ M$

(II) $\left[Ag^+\right] = \sqrt{k_{sp_{(AgI)}}} = 10^{-8}\ M$

(III) $Ag^+(aq.) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq); k_f$

$$k_f = \frac{\left[Ag(NH_3)_2^+\right]}{\left[Ag^+\right]\left[NH_3\right]^2}$$

$$10^8 = \frac{1}{\left[Ag^+\right](0.1)^2} \Rightarrow \left[Ag^+\right] = 10^{-10}\ M$$

(IV) $Ag^+(aq.) + 2CN^-(aq.) \rightleftharpoons Ag(CN)_2^-(aq.); k_f$

$$k_f = \frac{\left[Ag(CN)_2^-\right]}{\left[Ag^+\right]\left[CN^-\right]^2}$$

$$10^{11} = \frac{1}{\left[Ag^+\right](0.1)^2}$$

$$\left[Ag^+\right] = 10^{-13}\ M$$

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7. Ans. (D)

Sol. $PO_4^{3-} + H_2PO_4^- \longrightarrow 2HPO_4^{2-}$

5 2.5 –

2.5 x 5

$$pH = pKa_3 + \log\frac{\left[PO_4^{3-}\right]}{\left[HPO_4^{2-}\right]}$$

$$= 12 + 1\log\left(\frac{2.5}{5}\right) = 12 - \log 2$$

8. Ans. (B)

Sol. $$Ag^+ + 2CN^- \rightleftharpoons Ag(CN)_2^-;\ K_c = \frac{1}{4\times10^{-19}}$$

$$= \frac{1}{4}\times10^{19}$$

At t = 0; 0.03 0.1 –

At equilibrium ~ (0.1 – 0.06) 0.03

= 0.04

$$\frac{1}{4}\times10^{19} = \frac{(0.03)}{\left[Ag^+\right](0.04)^2}$$

$$\left[Ag^+\right] = 7.5\times10^{-18}\ M$$

9. Ans. (A)

Sol. Rate of hydrolysis; $r \alpha \left[H^+\right]$

$r = k[H^+]$ [ester]

$$\frac{1}{100} = \frac{\left[H^+\right]_{HA}}{\left[H^+\right]_{HX}} = \frac{\left[H^+\right]_{HA}}{1}$$

$$\left[H^+\right] = 10^{-2}\ M$$

For weak acid, $k_{a(HA)} = \frac{\left[H^+\right]^2}{C - \left[H^+\right]} = \frac{(10^{-2})^2}{1-10^{-2}} = 10^{-4}$

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10. Ans. (C)

Sol. $\Delta G^{o} = -RT\ell n\, k_p$

$-1743 = -8.3 \times 300\ \ell n\, k_p$

$\ell n\, k_p = 0.7$

$k_p = 2\ ; \qquad k_p = k_c (RT)^{\Delta ng}$

(A) $2 = k_c (0.082 \times 300)^{-1}$

$k_c = 2 \times (0.02 \times 300) = 49.2$

But from given data

$k_c = \dfrac{3}{3 \times 6} = \dfrac{1}{6}$

Both k_c values are not same

(B) $k_c = \dfrac{2}{(0.082 \times 300)^2} \neq \dfrac{6 \times 3^2}{3}$

(C) $k_c = 2(R \times 300)^{o} = \dfrac{6 \times 3}{(3)^2}$

Both k_c data are same.

(D) $k_c = 2 \neq \dfrac{(3)^2}{3 \times 6}$

11. Ans. (A,B,C)

Sol. (A) W.B. + salt of W.B. & S.A. $\Rightarrow$ buffer

(B) $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

2	1	–
1	x	1

W.A. + salf of W.A & S.B. $\Rightarrow$ Buffer

(C) $NH_3 + HCl \longrightarrow NH_4Cl$

2	1	–
1	x	1

W.B. + salt of W.B. & S.A. $\Rightarrow$ Buffer

(D) $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

1	2	–
x	1	2

$\Rightarrow$ not buffer

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12. Ans. (B,D)

Sol. (A) $k_{sp} = [Fe^{+3}][OH^-]^3 \Rightarrow [Fe^{+3}] \propto \frac{1}{[OH^-]^3}$

$\Rightarrow [H^+]\downarrow \Rightarrow [OH^-]\uparrow \Rightarrow [Fe^{+3}]\downarrow$

(B) $pOH = pK_b + \log\frac{[NH_4Cl]}{[NH_4OH]}$

$[NH_4Cl]\uparrow \Rightarrow pOH\uparrow \Rightarrow pH\downarrow$

(C) $NH_4I \rightarrow$ Acidic (salt of W.B. & S.A.)
$Cr(NO_3)_3 \rightarrow$ Acidic (salt of W.B. & S.A.)
$KCN \rightarrow$ Neutral (salt of S.A. & S.B.)

(D) Degree of hydrolysis for salts of W.A. & W.B. is independent of their concentration.

13. Ans. (B)

Sol. (A) $CH_3COOH \Rightarrow pH = \frac{1}{2}pka - \frac{1}{2}\log C$

on adding NaOH

$$\underbrace{CH_3COOH + \underset{(L.R.)}{NaOH} \longrightarrow CH_3COONa + H_2O}_{\text{Buffer}}$$

(B) $0.1M\ NaHCO_3 \longrightarrow pH = \frac{pka_1 + pka_2}{2}$

(C) NH_3 increase solubility of AgCl by complex formation.

(D) $NH_3 + HCl \longrightarrow NH_4Cl$

(L.R.)

✔ × ✔

form buffer

14. Ans. (C)

Sol. $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

3 1 –

2 × 1

$pH = pka + \log\frac{s}{a} = pka + \log\left(\frac{1}{2}\right)$

$= (5 - \log 2) - \log 2 = 4.4$

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15. Ans. (A)

Sol. $\left[H^+\right]=\sqrt{ka_1+Ka_2}=10^{-9}; \left[OH^-\right]=\frac{10^{-14}}{10^{-9}}=10^{-5}\ mol/L$

INTEGER TYPE

1. Ans. (1.00)

Sol. $HCl + NaOH \longrightarrow NaCl + H_2O$ (First reaction occur between S.A. & S.B.)

m moles	2	1	–
	1	x	1

Final concentration after mixing of solutions.

$\left[HA\right]=\frac{10\times0.1}{30}=\frac{1}{30}M$

$\left[HCl\right]_{left}=\frac{1}{30}M$

$HA(aq.) \rightleftharpoons H^+(aq.) + A^-_{(aq)}\ ;\ k_a=10^{-5}$

$\frac{1}{30}$ $\frac{1}{30}$ –

$\left(\frac{1}{30}-x\right)\simeq\frac{1}{30}$ $\left(\frac{1}{30}+x\right)\simeq\frac{1}{30}$ x

(as $\frac{Ka}{C}<10^{-3}\Rightarrow$ x is negligible compare to $\frac{1}{30}$)

$K_a=\frac{[H^+][A^-]}{[HA^-]}=\frac{\frac{1}{30}\times[A^-]}{\frac{1}{30}}$

$\left[A^-\right]=K_a=10^{-5}M$

$=1\times10^{-5}M$

$\boxed{x=1}$

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2. Ans. (3.20)

Sol. $k_{sp} = 4s^3 = 4\times\left(2\times10^{-6}\right)^3 = 32\times10^{-18}$

In buffer solution. (solubility = x mol/L)

$Pb(OH)_2(s) \rightleftharpoons Pb^{+2}(aq) + 2OH^-_{(aq)};\ k_{sp}$

x $\quad (2x + 10^{-6})$

↗

due to buffer

$\simeq 10^{-6}$

$K_{sp} = \left[Pb^{+2}\right]\left[OH^-\right]^2$

$32\times10^{-18} = x\left(10^{-6}\right)^2$

$x = 3.2\times10^{-5}\ M$

Buffer does not allow $[OH^-]$ to change.

3. Ans. (1.00)

Sol. $P_{He(I)} = P_{He(II)} = 2\,atm$

I chamber

$4 = \cancel{P_{NH_3}} + \cancel{P_{H_2S}} + \cancel{P_{He(I)}}$

P P

$\Rightarrow P = 1\,atm$

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2s(g);\ kp$

P P

$kp = 1\times1 = 1$

4. Ans. (4.00)

Sol. At STP $\Rightarrow v \propto n$ (same P & T)

$2COF_2(g) \rightleftharpoons CO_2(g) + CF_4(g)$

Ratio of moles at equilibrium	100	200	200
Partial pressure ($P \propto n$)	2	4	4

$V_{CF_4} = 500 - 300 = 200\,ml$

$\Rightarrow k_p = \dfrac{4\times4}{2^2} = 4$

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